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Practical No. 2

Aim: Smart City Sensor

- Scenario: A smart city project requires calculating the area and perimeter of circular parks.

Task: Write a program using Math.PI to calculate area/perimeter. If radius ≤ 0, display "please enter non zero positive number".

Name of Student:

Roll No.:

Semester/Year:

Academic Session:

Date of Performance:

Date of Submission:

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Objective

- To understand and use the built-in Math.PI constant in Java.
- To apply the mathematical formulas for area and circumference of a circle.
- To implement input validation using conditional (if-else) statements.
- To handle invalid/edge-case input gracefully with a meaningful message.
- To take user input in Java using the Scanner class.

Theory

The Math class in java.lang provides mathematical constants and methods that do not need an object to be created (all members are static). Math.PI is a static, final (constant), double-type field that returns the value of pi ($\pi \approx 3.14159265358979$), the ratio of a circle's circumference to its diameter.

For a circle of radius r, the standard formulas are:

Area = $\pi \times r^2$

Perimeter (Circumference) = $2 \times \pi \times r$

Since a radius cannot logically be zero or negative, the program must first validate the input using an if-else construct. If the radius entered is less than or equal to 0, the program should not proceed with the calculation and must instead display the message: "please enter non zero positive number".

Algorithm

Step 1: Start.
Step 2: Import the Scanner class (java.util.Scanner) to read input.
Step 3: Input the radius of the circular park (as a double).
Step 4: Check if radius ≤ 0.
Step 5: If true, display "please enter non zero positive number" and stop.
Step 6: If false, calculate area = Math.PI * radius * radius.
Step 7: Calculate perimeter = 2 * Math.PI * radius.
Step 8: Display the area and perimeter, rounded to 2 decimal places.
Step 9: Stop.

Flow Logic (Pseudocode)

```
BEGIN
 READ radius
 IF radius <= 0 THEN
    PRINT "please enter non zero positive number"
 ELSE
    area <- Math.PI * radius * radius
    perimeter <- 2 * Math.PI * radius
```

```
    PRINT area, perimeter
  END IF
END
```

Result

The Java program was successfully written, compiled, and executed to calculate the area and perimeter of a circular park using Math.PI. The program correctly validates the radius input using an if-else statement and displays the message "please enter non zero positive number" whenever the radius is zero or negative, ensuring the smart city sensor system does not process invalid geometric data.

Viva Voice Questions

1. What is the value returned by Math.PI in Java, and which class is it defined in?
2. Why is Math.PI declared as static and final in the Math class?
3. What is the difference between area and perimeter (circumference) of a circle?
4. What is the purpose of the Scanner class in Java? Name another way to take input.
5. Why is input validation important in real-world sensor-based systems?
6. What exception is thrown if a non-numeric value is entered where nextDouble() is expected, and how can it be handled?
7. What is the difference between System.out.print, println, and printf?